SANAAD سند

Solution Architecture Document

Re-Engineering from Broadcom Layer7 / Kony to Flutter + Node.js

Client: Hamad Medical Corporation (HMC)

Project Codename: SANAAD-MOBILE-APP

Version 0.1 | March 2026

Status: Draft for Review

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1 Introduction

This document describes the solution architecture for SANAD Mobile Application, the document covers the key architectural aspects from solutioning and technical perspectives, the document aims to document all the solution components that will be built during the implementation phase.

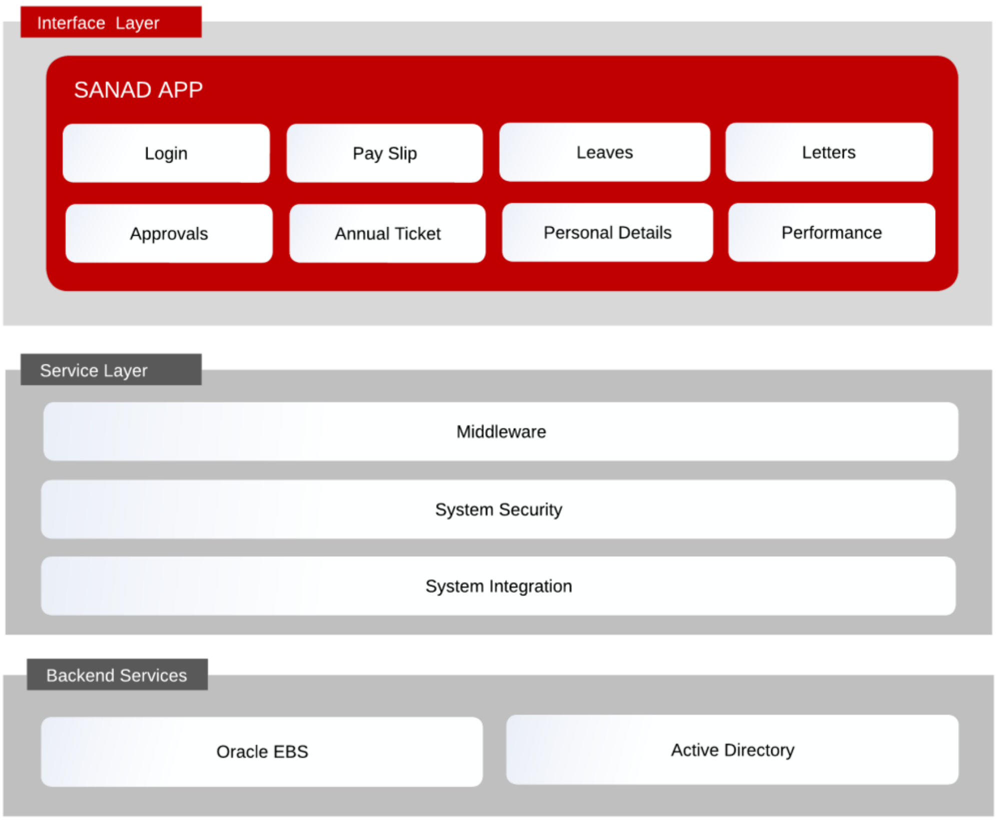
1. Document Audience

   This document is intended to be used by the following teams and stakeholders:

   a. malomatia functional and technical Project Teams
   b. HMC functional and technical Project Teams
   c. Program management team

2 Solution Overview

2. Logical Architecture



3 Solution Architecture

3.1 Hardware

3.1.1 Development Environment

VM Name	Description	Processor	Memory	Storage
mss-dev-web1	API Gateway 1	2	8 GB	50 GB
mss-dev-app1	Microservices 1	2	8 GB	50 GB

3.1.2 Staging Environment

VM Name	Description	Processor	Memory	Storage
mss-stg-web1	API Gateway 1	4	8 GB	90 GB
mss-stg-web2	API Gateway 2	4	8 GB	90 GB
mss-stg-app1	Microservices 1	4	8 GB	90 GB
mss-stg-app2	Microservices 2	4	8 GB	90 GB

3.1.3 Production Environment

VM Name	Description	Processor	Memory	Storage
mss-prd-web1	API Gateway 1	8	16 GB	150 GB
mss-prd-web2	API Gateway 2	8	16 GB	150 GB
mss-prd-app1	Microservices 1	8	16 GB	150 GB
mss-prd-app2	Microservices 2	8	16 GB	150 GB

3.2 Communication matrix

3.2.1 Development

URL: http://devmobilews.hmc.gov.qa

Source	Destination	Ports
mss-dev-web1	mss-dev-app1, mss-dev-app2	5001, 5002
mss-dev-app1	ldaps://hmc.gov.qa	636
mss-dev-app1	Oracle DB	1521

3.2.2 Staging

URL: https://stgmobilews.hmc.gov.qa

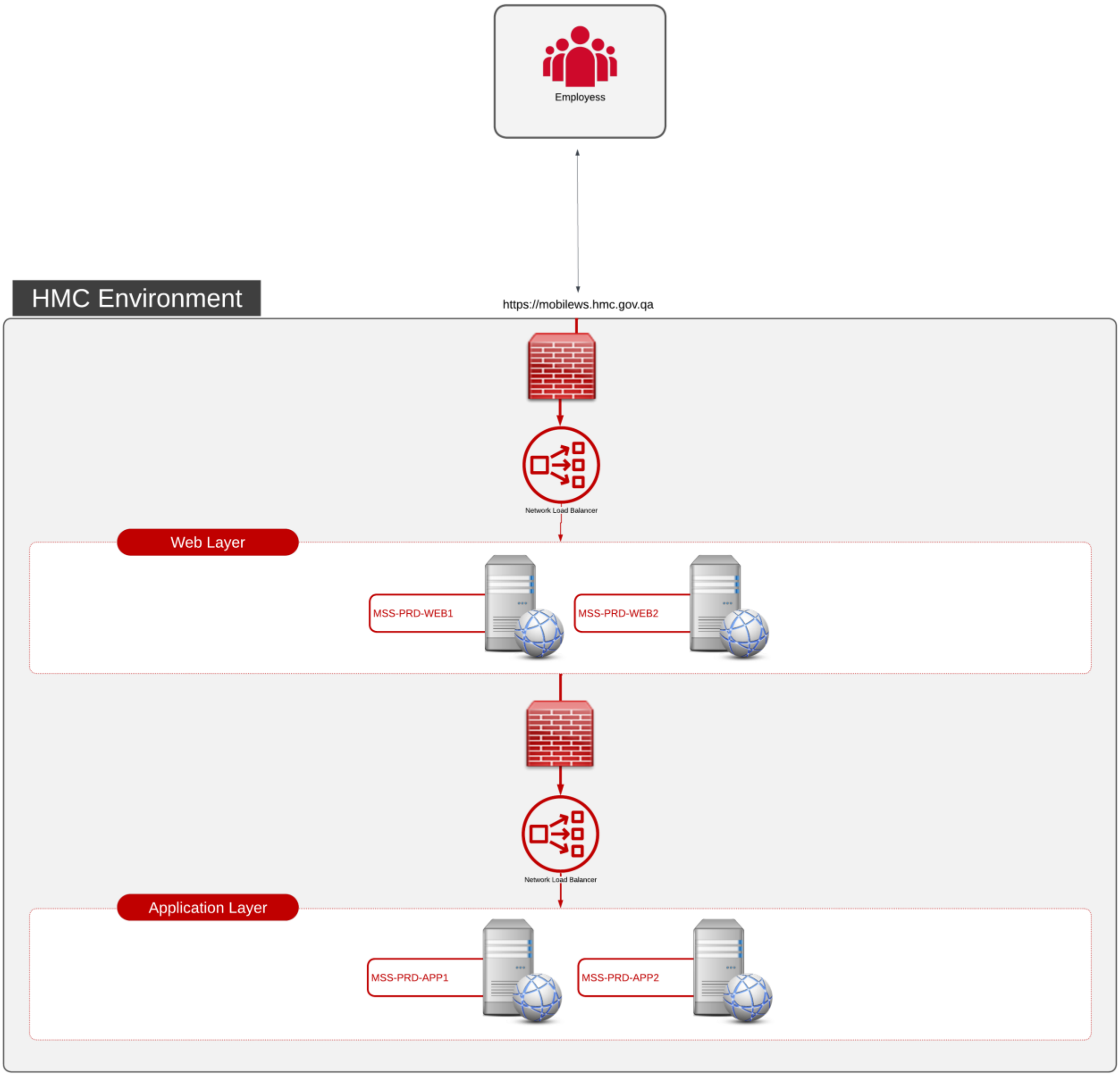
Source	Destination	Ports
mss-stg-web1	LB (mss-stg-app1, mss-stg-app2)	5001,5002
mss-stg-web2	LB (mss-stg-app1, mss-stg-app2)	5001,5002
mss-stg-app1	ldaps://hmc.gov.qa	636
mss-stg-app2	ldaps://hmc.gov.qa	636
mss-stg-app1	Oracle DB	1521
mss-stg-app2	Oracle DB	1521

3.2.3 Production

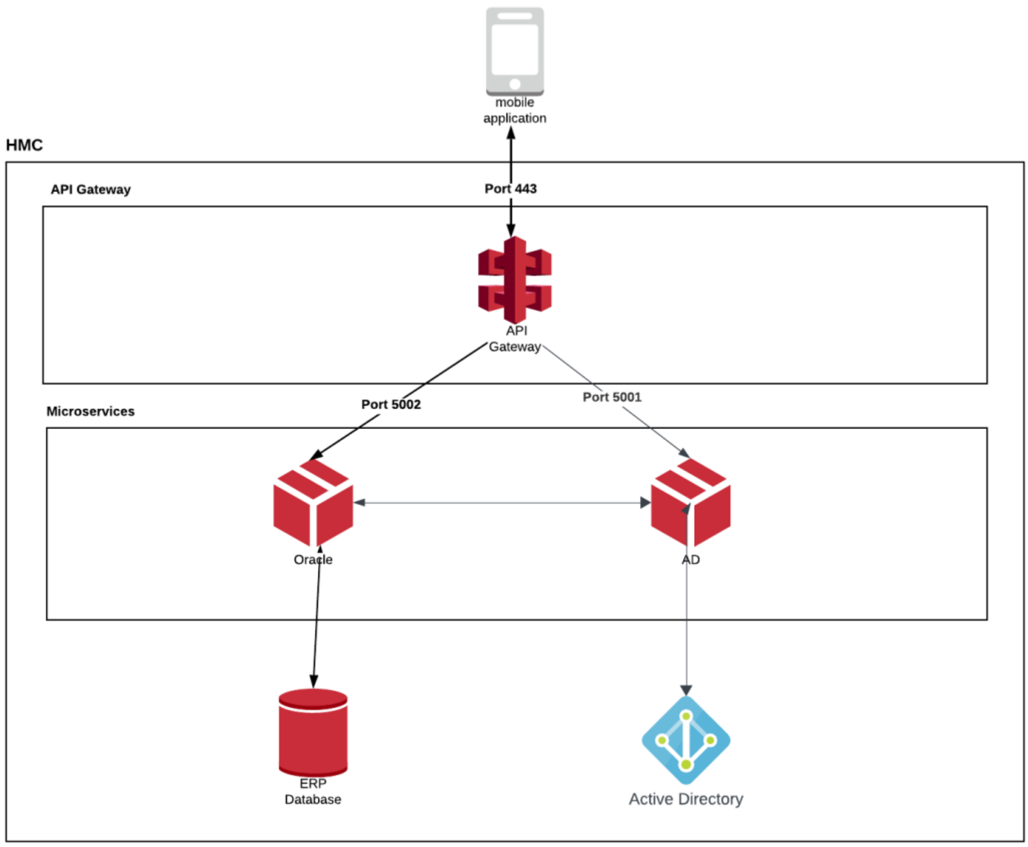
URL: https://mobilews.hmc.gov.qa

Source	Destination	Ports
mss-prd-web1	LB (mss-prd-app1, mss-prd-app2)	5001, 5002
mss-prd-web2	LB (mss-prd-app1, mss-prd-app2)	5001, 5002
mss-prd-app1	ldaps://hmc.gov.qa	636
mss-prd-app2	ldaps://hmc.gov.qa	636
mss-prd-app1	Oracle DB	1528
mss-prd-app2	Oracle DB	1528

3.3 Deployment architecture



3.4 Microservices architecture



The gateway receives the request and forward it to appropriate microservice.

3.5 Security Measures

- Architecture
  - Web layer is hosted in DMZ and separated from App Layer
- HTTP Requests
  - All requests between mobile application and servers are over HTTPS
  - SSL Pinning is implemented
  - As an extra encryption layer, all requests body are encrypted (AES-256-CBC)
- Authentication:
  - Two factor authentication is implemented (OTP via SMS)
  - Every request is verified by checking against secure secret key (JWT)
- Authorization:
  - Every request is verified by checking the token of the user verse the requested data
- User token get expired after 30 minutes of inactivity (Configurable)